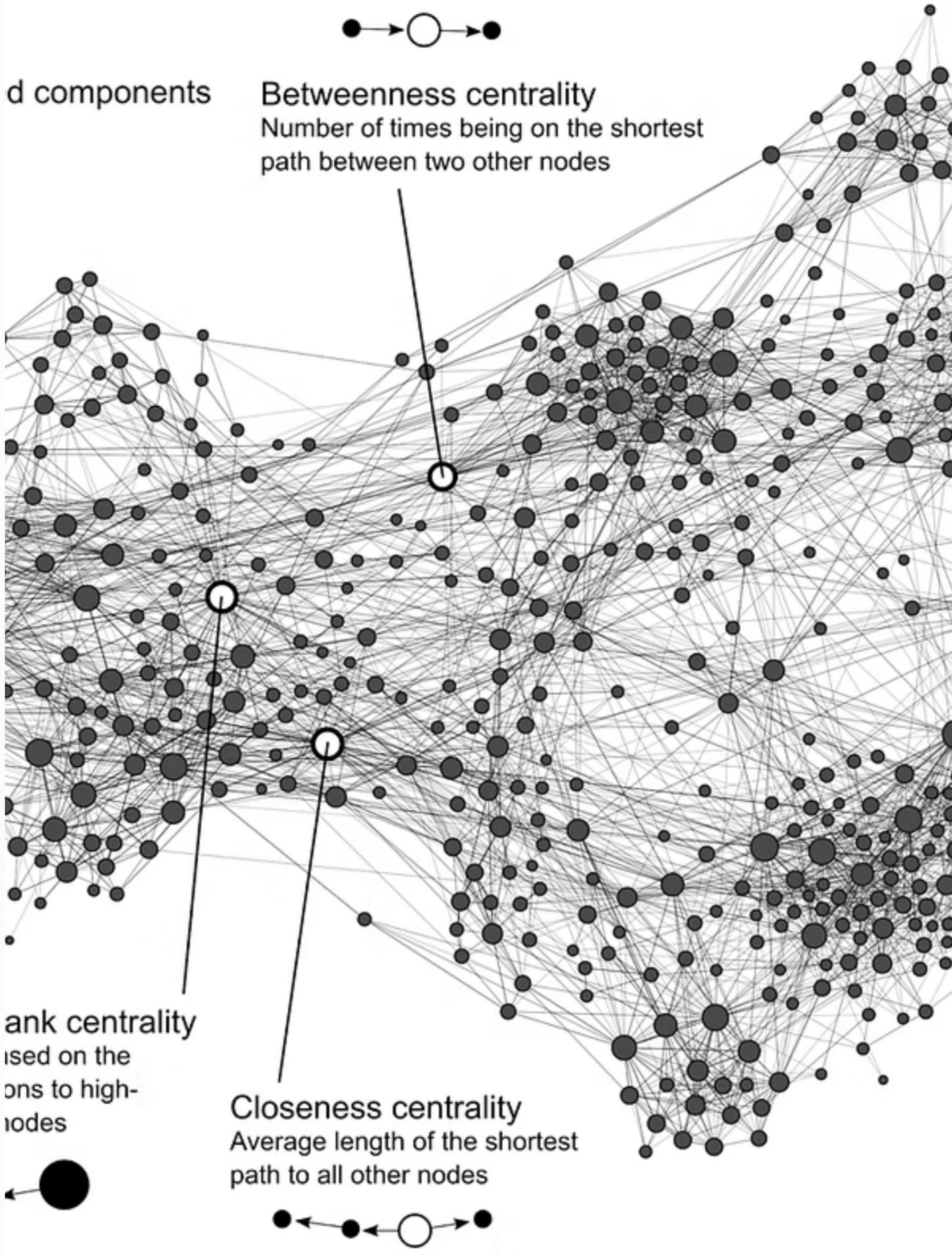
Variational Autoencoder (VAE) for Image Generation

A probabilistic approach to generative modelling that learns to create realistic images by understanding the underlying data distribution. Built with PyTorch for robust implementation and scalability.

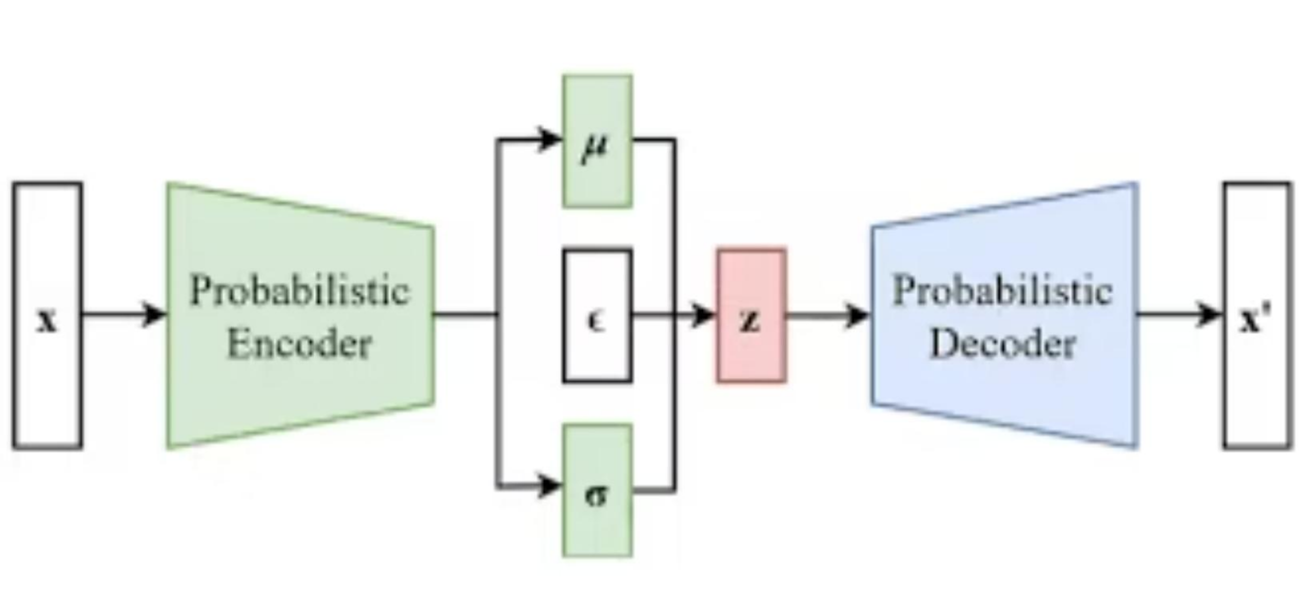


What is a Variational Autoencoder?

Beyond Standard Autoencoders

A VAE extends traditional autoencoders by learning a probability distribution over the latent space, rather than deterministic encodings. This probabilistic framework enables the model to generate entirely new samples that haven't been seen during training.

The key innovation lies in treating the latent representation as a distribution, allowing for smooth interpolation and controlled generation of novel outputs.



Architecture Overview

Encoder Network

Maps input images to latent distribution parameters: mean (μ) and variance (σ^2)

Reparameterisation Trick

Samples from $N(\mu, \sigma^2)$ whilst maintaining differentiability for backpropagation

Decoder Network

Reconstructs images from sampled latent vectors, learning generative mapping

Mathematical Foundation

Reparameterisation Trick

$$z = \mu + \sigma \odot \epsilon$$

This formulation allows gradients to flow through the sampling operation, making the model trainable via backpropagation.

Loss Function

Reconstruction Loss: Measures how well the decoder recreates the input

KL Divergence: Ensures latent distribution stays close to prior N(0, I)

Implementation Specifications

Optimisation

Adam Optimizer

Learning rate: 1e-3

Beta values: (0.9, 0.999)

Architecture

Latent Dimension: 64

Convolutional encoder/decoder

Batch normalisation layers

Framework

PyTorch

GPU-accelerated training

Modular architecture

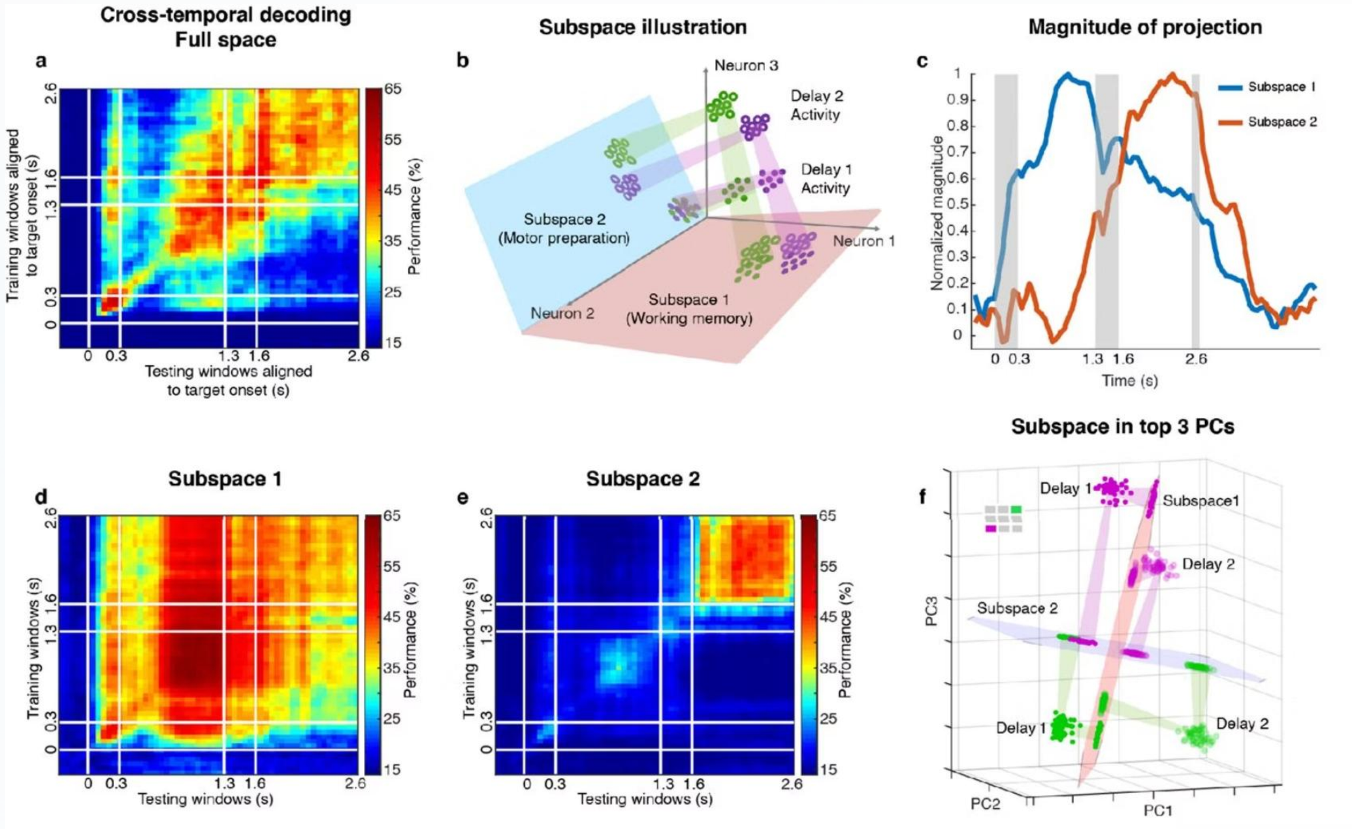
Experimental Results

The trained VAE demonstrates strong performance in both reconstruction accuracy and generation quality. The model successfully captures the underlying data distribution of CIFAR-10 images.

Image Reconstruction

High-fidelity reconstructions with preserved semantic features and colour distributions

Latent Space Interpolation



Smooth transitions between images reveal structured, continuous latent representations

Real-World Applications

Image Synthesis

Generate diverse, realistic images for data augmentation, content creation, and synthetic dataset generation. Particularly valuable when real data is scarce or expensive to obtain.

Image Denoising

Remove noise and artifacts from corrupted images by leveraging the learnt clean data distribution. Effective for medical imaging, satellite imagery, and historical photo restoration.

Anomaly Detection

Identify outliers by measuring reconstruction error—high error indicates samples that deviate from learnt distribution. Applications in quality control, fraud detection, and medical diagnostics.

Key Takeaways

Efficient Latent Representations

VAEs learn compact, meaningful representations that capture essential data structure whilst enabling smooth interpolation in latent space.

Generative Capabilities

The probabilistic framework allows generation of realistic variations and novel samples, going beyond simple memorisation of training data.

Foundation for Advanced Advanced Models

VAE principles underpin modern generative architectures including VAE, VAE-GAN models that push the boundaries of image synthesis.